Pulse Count Rate Measurements and Defect Correlation of Veto Photon-Detector Units for the DarkSide-20k Experiment

Abstract

Tile characterisation is part of the production QA/QC of Photo-Detector Units (PDUs) for the DarkSide-20k dark matter search experiment. Failing tiles are inspected for defects, with mechanical damage and electrical failures found to be indicators of elevated Pulse Count Rate (PCR) in $60-70\%$ of cases. The intrinsic Dark Count Rate (DCR) of SiPM detectors differs from PCR in production test stands, by electronic noise pedestal leakage in the single photo-electron counting window and residual photon backgrounds. PDU PCR can be predicted from individual tile measurements, with discrepancies explained by these effects. A Gaussian Subtraction counting method was developed and implemented in the characterisation to remove baseline noise leakage from test data, correctly reducing the measured quadrant PCR by 2.7× for Liverpool runs and 1.4× for AstroCeNT, as well as the difference between quadrant and tile-average values, from 3.2× to 1.4× and from 2.4× to 1.2× respectively. An internal dark box was deployed to remove light backgrounds from Oxford PDU test data, yielding average corrected values of $(0.016 \pm 0.008)\,\mathrm{Hz/mm^2}$ for a normal quadrant and $(0.013 \pm 0.008)\,\mathrm{Hz/mm^2}$ for one containing higher-PCR tiles, along with a $1.5-2\times$ excess to the full quadrant measurement.

Keywords: DCR, PCR, Photo-Detector Unit, Gaussian Subtraction, signal addition, defect correlation

1 Introduction

The nature of dark matter (DM) is one of the most compelling open problems in modern physics. Astrophysical evidence ranging from galactic rotation curves and gravitational lensing, to large-scale structure formation and CMB measurements, have long established that approximately 27% of the energy content of the Universe is comprised of non-baryonic matter that does not interact with light [1]– yet its fundamental essence is still unknown.

Many theoretical candidates have been proposed, notably axions, sterile neutrinos, primordial black holes, as well as Weakly Interacting Massive Particles (WIMPs), which remain among the most theoretically motivated [2]. Their scientific foundation arises from considerations of supersymmetry and predicts a relic abundance consistent with observations.

In theory, WIMPs may be detected through their extremely rare elastic scattering off atomic nuclei. Such an interaction would produce low energy nuclear recoils, and generate scintillation light and ionisation electrons in a detector medium, which can be measured. This *target* medium should thus be aimed at having high scintillation and ionisation yields, but also the possibility to clearly distinguish between nuclear and electron recoils[1].

Liquid argon (LAr) is a particularly suitable target [3] and, in conjunction with the dual-channel readout technique, forms the basis of the DarkSide-20k experiment.

1.1 The DarkSide-20k Experiment

DarkSide-20k (DS-20k) is a dual-phase LAr Time Projection Chamber (TPC) under construction at the INFN LNGS laboratory in Italy, estimated to start science operations from 2028. DarkSide will search for a wide range of dark matter candidates from keV to Planck mass scales, including WIMPs; the detector is designed to have less than 1 instrumental background event in 10 years of operation in the WIMP search.

The detector (Figure 1) consists of three surrounding volumes. The innermost is the TPC, containing around 50 tons of underground argon (UAr), extracted from deep wells and thus thoroughly depleted of radioactive isotopes. DM interactions produce a rapid scintillation signal (S1) from prompt photons generated by dark matter-induced recoils, and a delayed ionisation signal (S2) from electrons drifting upward in the induced electric field and generating electroluminescence photons. Around the TPC, there is an inner veto still of UAr, followed by an outer veto of atmospheric argon (AAr). These protect the target against neutrons and cosmic muons respectively and, together with the TPC, are all equipped with ultra-sensitive photon detectors, called Silicon Photomultipliers (SiPMs).

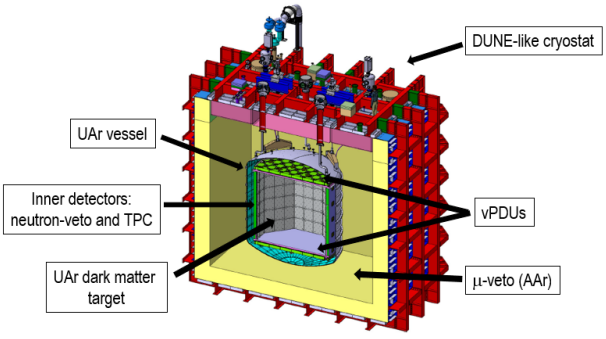


Figure 1: Cross-section of the DS-20k detector at LNGS [4].

[1]Gamma rays and beta electrons produce electron recoils which are much more abundant than the WIMP-induced nuclear recoils.

1.2 Detector Specifications and Characterisation

A SiPM is a solid-state detector incorporating numerous Single Photon Avalanche Diodes (SPADs) operated in Geiger mode. SiPMs are designed to detect individual incident photons with high quantum efficiency. When a photon is absorbed by a SPAD, it generates an electron-hole pair that triggers a self-sustaining charge avalanche, producing a current pulse. The probability of this occurring for an incident photon is given by the Photon Detection Efficiency (PDE), which for the NUV-HD-Cryo devices of DarkSide-20k reaches values of $\sim 50\%$ at LAr emission wavelengths. In contrast with conventional photomultiplier tubes, SiPMs have lower operating voltage, negligible radioactive contamination, and are capable of operating at cryogenic temperatures [5].

Figure 2 illustrates how 24 SiPMs are mounted to a PCB, forming a *tile* of $\approx 22.4\,\mathrm{cm}^2$ active area. This PCB provides a common high voltage (HV) bias to all SiPMs via a voltage divider, and also contains a cryogenic transimpedance amplifier powered by a low voltage supply (LV), which delivers the summed current from the 24 cells as a single output channel through a 6-pin connector. In DarkSide-20k's readout, the Photon Detector Unit (*PDU*) is a 20 cm x 20 cm object composed of 16 tiles mounted on a motherboard PCB that distributes voltage and control signals, and sums groups of tiles. Four tiles with a shared readout channel form a *quadrant*, and four quadrants attached to a motherboard form the full vPDU (when referring to the veto).

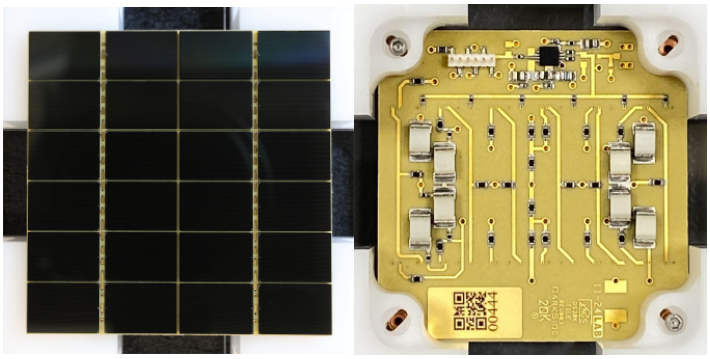


Figure 2: (Left) One detector tile, made up of 24 SiPMs, as seen from the top. (Right) Bottom view of the PCB connected to it: the 6 pin connector is visible in the top-left, with pins for ground, output, LV-, LV+, power down and HV bias. (Images taken from [6])

Before installation, each tile must undergo a careful quality protocol: at the NOA facility at LNGS for the 528 TPC PDUs, and at a consortium of ten UK groups plus the AstroCeNT institute in Poland for the 120 vPDUs respectively. This involves warm (room temperature) and cold tests (in liquid nitrogen at 77 K), in order to evaluate the tiles and units against production stability criteria. Such tests evaluate the IV curve breakdown voltage, noise spectrum and photon-detection signal and noise characteristics in pulsed-laser data. The latter simulates photon veto interactions and gives rise to the so-called *finger plots* – amplitude histograms displaying discrete photo-electron (PE) peaks (see Figure 7). Optical inspection in cleanroom environments is conducted at all stages (SiPMs, tiles, PDUs) testing for surface defects and voltage draw imperfections, successively rejecting damaged components [6].

vPDU tests in particular are conducted across multiple facilities in Oxford, Edinburgh, Liverpool, and AstroCeNT in Warsaw (as well as Manchester and Warwick, not considered in our analysis). Given their parallel operations to monitor production in near real time, systematic verification and benchmarking across datasets is done to ensure consistency.

1.3 Dark Noise

SiPMs are subject to various intrinsic sources of noise, independent of incident photons. One such source is the dark current rate (DCR), where thermally excited charge carriers spontaneously trigger an avalanche in a SPAD, producing a pulse identical in shape with that of a photon-induced single photo-electron (SPE). Dark counts are the primary source of noise in the detectors, along with noise pulses arising from optical crosstalk (caused by a primary avalanche emitting secondary photons that trigger neighbouring SPADs), as well as afterpulses (coming from delayed charge carriers initially trapped when in avalanche) [7].

DCR is optimally measured in vacuum, at single device level, which is what was done to select and characterise the SiPM technology for DarkSide-20k. However, the production test stands operate tiles and PDUs (made of 24 and 96 SiPMs summed per channel each) submerged in liquid nitrogen (LN2), to mimic their operating conditions in LAr. As a result, there are additional electronics noise and spurious light contributions to the uncorrelated noise rate (which in vacuum for a single SiPM is the DCR). The test stands make a simple and robust measurement of the uncorrelated pulse count rate (PCR), which has contributions from DCR as well as these additional sources. This project focuses on understanding the relationship between the measured PCR and the intrinsic DCR.

The aim of this project is to test tiles and vPDUs, and to study in detail the consistency between DCR measured at tile and PDU level. We begin by describing the experimental setup and the methods used in conducting our post-QA/QC tests. Then we continue with the analysis performed on defective tiles and on the DN Rates of characterised vPDUs. Finally we describe discrepancies in the initial results, corrected statistics, signal addition and a vPDU dark box study.

2 Methods and Setup

We conduct the characterisation of tiles and vPDUs at the University of Oxford across two laboratory settings.

2.1 Cleanroom Tile Inspection

The process commences in an ISO 7 cleanroom environment, where detectors are tested at a single-tile level using a cryogenic test stand, a CMOS imaging system and a digital microscope. Tiles are classified according to QA/QC criteria based on their electrical and signal performance, evaluating parameters such as the DCR, signal-to-noise ratio (SNR), and the shape of the IV profile, among others (see Appendix D).

Tiles that fail to meet these criteria are rejected and further inspected for anomalous features. One indicator of erroneous behaviour is the breakdown in the IV profile[2]. Tiles which are considered *good* (Grade A) present a single, well-defined breakdown (Figure 4), defective ones often describe double breakdown or none at all.

To diagnose which SiPM is problematic, the tile is biased above breakdown and imaged with a long-exposure by a low-noise CMOS camera. The tile is connected to bias voltage during the measurement, and images are acquired with a 600 s exposure inside a dark box, as ambient light would interfere with imaging. The CMOS technique aims to measure photons emitted by the avalanche process in the SiPMs.

Figure 3 shows an image of a tile taken with the CMOS; each numbered square is one SiPM. SiPMs that are brighter than their neighbours have higher overall photon emission, and highly unbalanced pairs (such as 11, 5), are characteristic of tiles that fail the IV curve QA/QC criterion because of having a double breakdown feature. Bright regions within a single SiPM, referred to as *hot spots*, indicate SPADs with higher emission, which may be correlated with visible surface damage to a SPAD, and potentially with higher PCR for that SiPM[3].

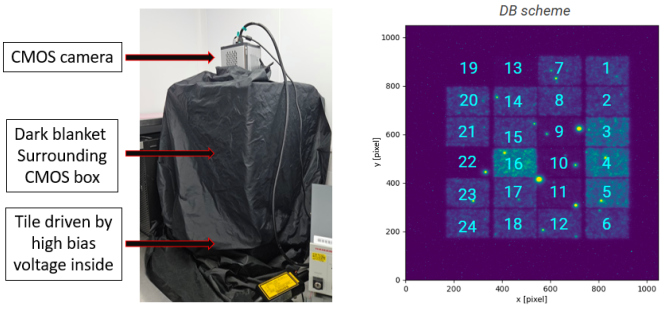


Figure 3: CMOS system and the image of a highly defective tile.

Once the electrical issue has been identified, we subject the tile to a further visual inspection meant to emphasize any small scale defects which can propagate to overall performance. Among others, scratches, surface depositions, rail damage, deficient wire bonds and dead cells can provide a direct link between physical defects and measured QA/QC failures. Such a performance prediction is possible given that only tiles passing QA/QC criteria are assembled into PDUs.

2.2 Tile Defects Correlation

The second feature we evaluate is possible correlations between tile defects and measured PCR. 39 tiles with identified defects from CMOS imaging and visual inspection conducted by us were characterised beforehand with the original PCR counting method in a dark box setup. Defects are grouped into broader classes mirroring their physical origin: **Electrical** (bad SPE, high RMS, extra peaks), **Mechanical** (scratches, deposits, dirt, broken rails, defective wire bonds), **Failure** (current limit, double breakdown, inactive tiles, no LV), and **Unknown** (no visible damage or defect identified). The findings are summarised in Table 1, with the full dataset provided in Appendix F.

Table 1: Summary of defect types and associated PCR (in Hz/mm^2). The mean and median PCR of tiles in each class is compared.

Defect	# tiles	Mean PCR	Median PCR
Electrical	12	0.16	0.10
Mechanical	13	0.17	0.11
Failure	11	0.30	0.11
Unknown	3	0.15	0.02

A clear trend is observed for Failure tiles: these have the largest PCR spread, with the highest observed rates, but also most suppressed values where tiles are virtually non-operational. This suggests that such defects tend to increase PCR on average, occasionally rendering tile behaviour unreliable altogether. Tiles with Electrical defects such as high RMS present elevated PCR values, with $60-70\%$ of them above the median of the entire dataset (not just of the class). In contrast, those with bad SPE and extra peaks show no consistent connection to elevated or reduced rates, as can be inspected in Appendix F. Mechanically damaged tiles are moderately linked to increased PCR, about 60% of them above the global median, but counterexamples do exist. We also note that the extent and location of the defects affect their impact on PCR values. Lastly, tiles with no visible damage still register elevated PCR, demonstrating that poor performance may not always be predicted by optical inspection alone. Importantly, given the limited sample size for this analysis, our results are to be interpreted as indicative trends rather than quantitative and predictive

[2]The IV profile we expect is not linear, but rather flat up to some voltage, having no counts thus minimal current, after which the avalanche triggers and the curve abruptly jumps at the breakdown point (Figure 4).

[3]It should be noted that hot spots can also be present in nominally functioning tiles.

conclusions. Nonetheless, this study is relevant as an intuitive framework for future installation, but it should be extended for higher-quality use and full PDU imaging.

2.3 vPDU Tests and Assembly

16 tiles passing their QA/QC are assembled onto one motherboard PCB to form a PDU (for a pre-production example, see Appendix A), then tested at room and cryogenic temperatures to characterise performance. The dewar setup used is illustrated in Figure 4, and a schematic of the PDU quadrants and connections in the test stand is shown in Figure 5. The vPDU is attached to supports and lowered into the dewar – this is originally empty for warm tests, then filled with LN2 for the cold tests. Its motherboard (MB) is connected with outgoing cables to the HV, LV and driven by a steering module, that enables and disables the signal output on each vTile or vMB quadrant. At the top and bottom of the dewar support, there are LED diffuser bulbs which send light from a laser into the closed environment. A digitizer reads the signal from the laser and from the vPDU and sends it to the computer. The vPDU outputs are also visualised on an oscilloscope to establish proper operation before acquisition.

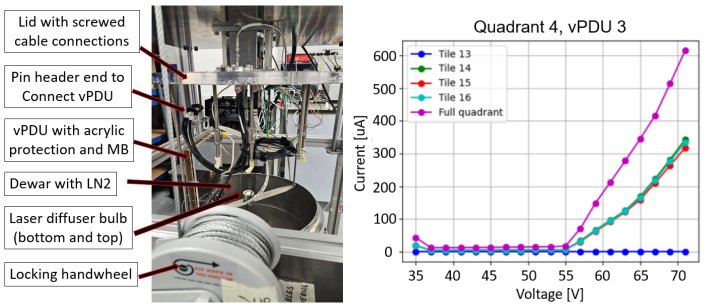


Figure 4: (Left) Cryogenic dewar setup: may be filled with LN2 and the lid closed to simulate dark tests; a dark box has also been implemented, surrounding the vPDU, and the dewar is further secured with gaffer tape for improved conditions. (Right) Characteristic IV curve of a given quadrant at cold: we note the breakdown voltage around 56 V for good tiles, with defective ones sometimes having double breakdown or no breakdown; we also note signal addition of individual tiles up to a quadrant, with tile 13 inactive in the given case.

Warm tests are intended for once again obtaining the IV curve, in this case for tiles and quadrants alike, as their signals are ultimately connected (Figure 4)[4]. Cold tests are devised such as to simulate the LAr environment the PDUs will experience in-situ in DarkSide-20k, and measure photon detection performance from single photons incident from a laser pulse. For this reason, avalanches are prompted by short laser pulses, and the signals recorded in 32 μs windows. Each waveform (registering voltage as a function of time, sampled at 250 MHz) contains both the laser-induced avalanche signal and the spontaneous uncorrelated counts occurring in the pre-trigger region, from which the PCR is measured and the DCR is estimated. Since we want to avoid any light leak that would produce further avalanches and inaccurately increase our rates, we isolate the dewar with gaffer tape, dark blankets and no lighting – we will refer to these as the *initial* tests. We subsequently encase the vPDU in an internal dark box, that shields the PDU from light produced in the LN2, as well for our *final* tests[5]. We search for signals from the tiles and quadrants: many come from 1 PE, fewer from 2 PE etc., but the vast majority are 0 PE noise.

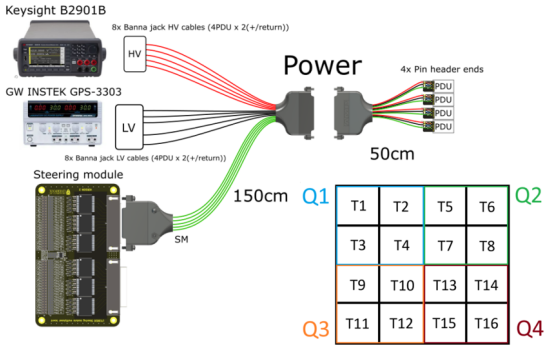


Figure 5: Electrical connections of the vPDU test stand. Four vPDUs may be studied simultaneously, we have analysed one in detail. Each unit has the layout presented, quadrants denoted by Q and tiles by T.

3 Data Analysis

3.1 Acquisition and Reconstruction

Sets of 600,000 waveforms are acquired for each combination of two quadrants or tiles on the available channels until we have statistics for the full vPDU. In each waveform, peaks in the signal are detected above a 3 RMS threshold from the baseline voltage, which correspond to avalanche pulses from one or more simultaneously firing SPADs, with their amplitude and timing recorded to reconstruct the PE spectrum. Figure 6 displays an example of the peak-finding procedure, with Appendix B containing the reconstruction code.

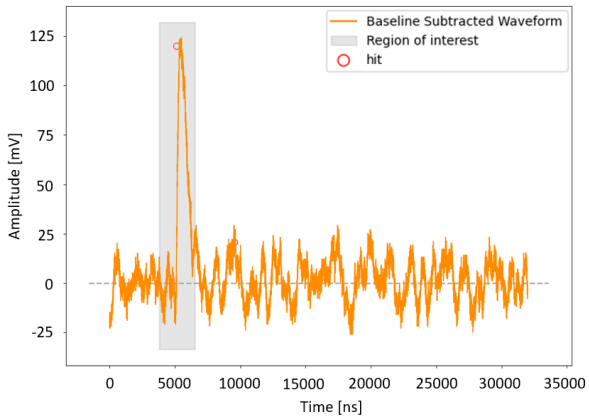


Figure 6: Sample of a waveform acquired in Oxford, with a clearly identified peak in the laser region of interest, highlighted in gray.

[4]For future reference, the tile and quadrant layout is shown in Figure 5.

[5]Note that the test stands are inherently not designed for dark tests, which are performed in vacuum to further eliminate Cherenkov radiation, in the construction process.

Figure 6 is an example of a voltage vs. time waveform. Samples on the x-axis are converted to time units by $t = N_{\text{samples}}/f_s$, where f_s is the sampling frequency – for our data, we had 4 ns/sample. The laser pulse arrives at sample ~ 1250 and lasts for ~ 100 samples. The timing range when the laser is on is termed the laser region of interest (ROI), and this region is used to calculate photon detection performance metrics. The region before the laser is on is termed the pre-trigger ROI and is used to calculate the uncorrelated noise pulse count rate.

For the automated analysis, the waveforms are reconstructed and processed by the *vPDUPassport* pipeline (Appendix B), henceforth referred to as *the passport*. It operates in either quadrant or tile mode, processes the ROOT files produced by the reconstruction code, and outputs the full set of QA/QC parameters: breakdown voltage, 1 PE amplitude, PCR, RMS and SNR. The passport first identifies the laser pulses, defining the laser ROI, then constructs the amplitude histogram of all peaks within that region. This is the finger plot and it represents the central diagnostic for our analysis.

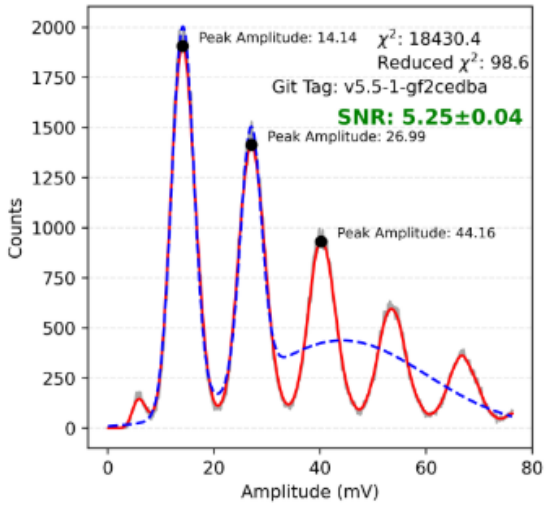


Figure 7: Full amplitude histogram (finger plot) in the laser ROI (1173 – 1373 samples $\Leftrightarrow$ 4.7 – 5.5μs) for Quadrant 1 of vPDU3.

One such finger plot we obtained for Q1 of vPDU3 is shown in Figure 7. It exhibits discrete peaks corresponding to integer multiples of the SPE amplitude, with the 1 PE peak around 14 mV, the 2 PE peak around 28 mV and so on. A smaller peak is observed close to zero amplitude: this is the residual of the pedestal, corresponding to noise (no SPAD firing). A multi-Gaussian fitting is then performed on the peaks, attempting to fit the first three by minimising the χ^2 value. It then identifies the peak amplitudes in counts, their positions and widths, and computes the SNR as:

$$\text{SNR} = \frac{\mu_2 - \mu_1}{\sigma_1}, \tag{1}$$

where μ_2 and μ_1 are the 2 PE and 1 PE positions (mV amplitudes) and σ_1 is the 1 PE standard deviation.

3.2 Dark Noise Counting

Once the 1 PE position and its standard deviation are found, the passport sets lower and upper thresholds at 2σ away to integrate 95% of the 1 PE peak. To isolate spontaneous dark counts from laser signals, we consider this 1 PE window exclusively in the pre-trigger region, more precisely, in the constant plateau dark noise ROI (897 – 1137 samples). The identified counts are divided by the total number of waveforms. This dimensionless dark noise rate is normalised by the time window lengths and active area, producing the PCR in Hz/mm^2. This procedure in practice is detailed below.

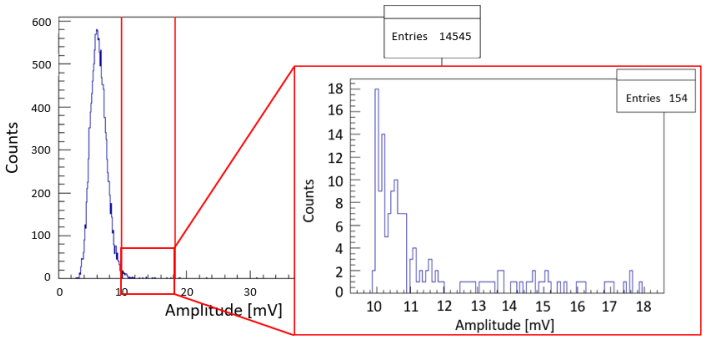


Figure 8: Histogram of counts in the DN ROI for vPDU3 Q1, with the 1 PE region zoomed in to highlight the number of DN counts = 154. The x-axis is in units of mV and the red lines indicate the 1 PE thresholds, given in ADC by the passport, and shown in mV on the plot using the 0.244 mV/ADC conversion rate.

Figure 8 illustrates the number of peaks in the DN ROI. Of that region, we are specifically interested in the number of pulses falling inside the 1 PE window, between 9.947 – 18.315 mV, which is $N_{\text{counts}} = 154$. The ROI time window is $t = 240$ samples $= 0.96\,\mu s$ and the active detection area $A = 4 \times 2240\,\text{mm}^2$, since the quadrant is made up of four active tiles. The calculation then follows:

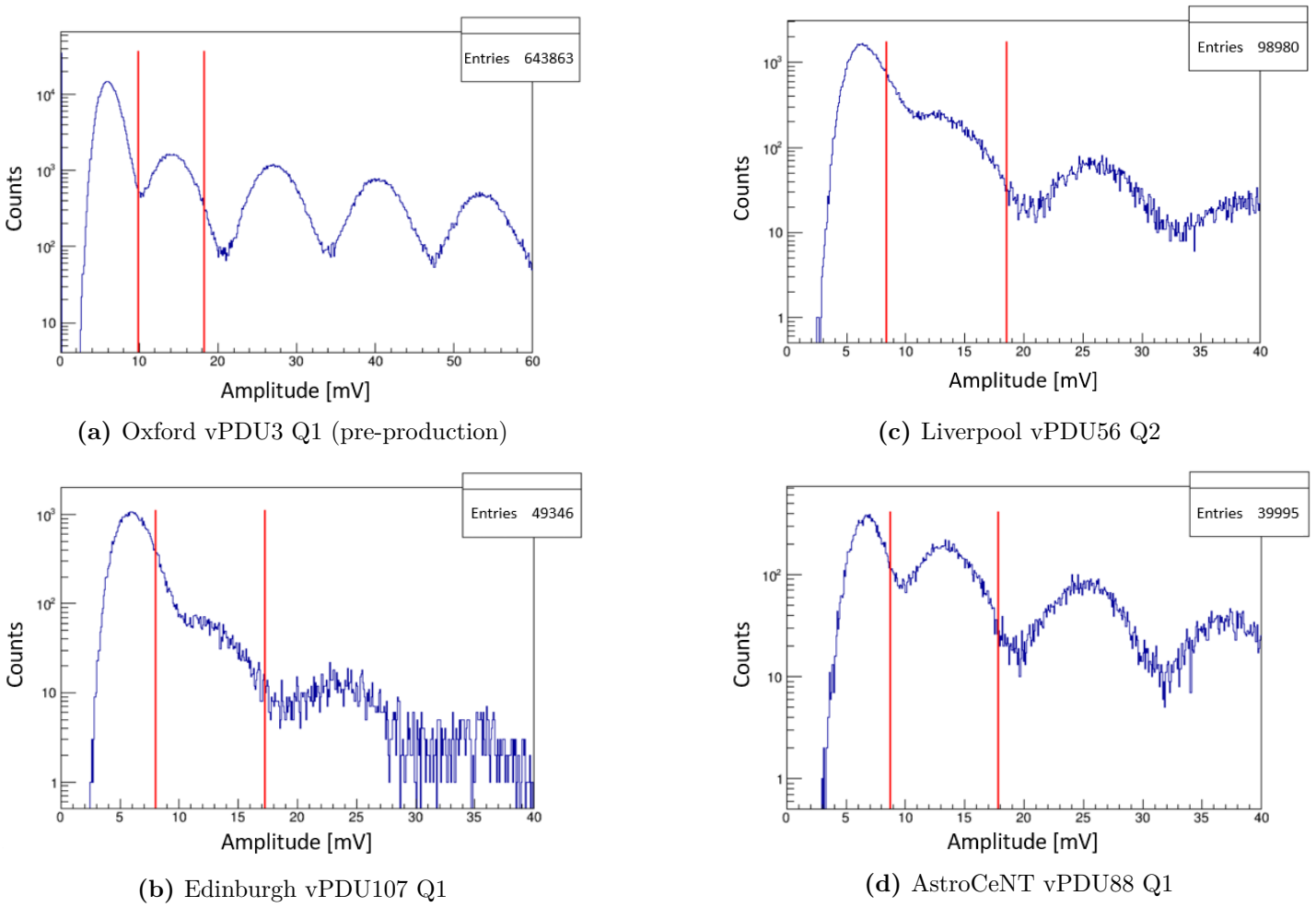
$$\text{DN Rate} = \frac{N_{\text{counts}}}{N_{\text{WF}}} = (5.02 \pm 0.44) \times 10^{-4}\,, \tag{2}$$

$$\text{PCR} = \frac{\text{DN Rate}}{t \times A} = (0.057 \pm 0.005)\,\text{Hz/mm}^2. \tag{3}$$

These manually calculated results match the outputs of the passport, which are provided in full for this test, and in the form of a table for the various tile combinations considered in Appendix D.

Beyond the PCR, the passport extracts two further correlated noise parameters for each acquisition. The Correlated Delayed Avalanche rate (CDA) gives the mean number of pulses in a post-trigger ROI, with the expected number of dark pulses subtracted. It thus captures correlated events like afterpulsing and delayed cross-talk that follow shortly after a primary avalanche.

$$N_{\text{CDA}} = \int_{t_1}^{t_2} [\,R(t) - R_{\text{DN}}(t)\,]\,dt \tag{4}$$



(a) Oxford vPDU3 Q1 (pre-production)

(c) Liverpool vPDU56 Q2

(b) Edinburgh vPDU107 Q1

(d) AstroCeNT vPDU88 Q1

Figure 9: Signal amplitude spectra of selected Grade A vPDU tile quadrants, from Oxford, Edinburgh, Liverpool and AstroCeNT. The plots correspond to the full spectrum (finger plot) on logarithmic scale, with the 1 PE window marked by the red lines. The 1 PE limits also differ at each facility, and the cruder aspect of plots other than Oxford is due to reduced acquired waveforms. The hardware differences in the LV supply of the pre-production vPDU from Oxford cause better separation compared with production ones.

Here, $R(t)$ is the total rate of pulses in the time window and $R_{\mathrm{DN}}(t)$ is the rate of dark noise pulses, with $t_1 = 1373$ and $t_2 = 2623$ samples defining the CDA ROI.

Lastly, the Normalised After-Pulse Activity (NAPA) quantifies similar correlated noise in the post-trigger region, where trapped carriers are released following the primary avalanche and trigger the same SPAD once again. This also considers Delayed Cross-Talk (DeCT) between the tiles, and is thus computed as the mean normalised pulse amplitude in the post-trigger region, of pulses with amplitudes above a 0.66 PE amplitude threshold, without the primary contribution.

$$\mathrm{NAPA} = \left\langle \frac{A_i}{A_{1\mathrm{PE}}} \right\rangle_{A_i/A_{1\mathrm{PE}} > 0.66} - 1 \qquad (5)$$

For the same quadrant as before, we obtain CDA $= 0.211 \pm 0.046$ and NAPA $= 0.573 \pm 0.189$.

The PCR and CDA results confirm 10× lower values than the respective means for Grade A tiles: $\overline{PCR} = 0.57\,\mathrm{Hz/mm^2}$, and $\overline{CDA} = 2.1$ (see Appendix D for a full account of QA/QC statistics across facilities and initial results).

Since each tile has independently passed production QA/QC with measured PCR values below the Grade A threshold, the quadrant PCR can in principle be predicted from the individual tile measurements. Assuming uncorrelated Poisson processes, the quadrant rate should equal the sum of the four tile rates, and the normalised Hz/mm^2 values should be consistent across hierarchies. For Q1 of vPDU3, the individual tile runs (see Appendix D) yield tile-level DN rates of $\sim 9 \times 10^{-5}$, which would sum to a predicted quadrant rate of $\sim 3.6 \times 10^{-4}$ that is below the measured 5.0×10^{-4}. Two effects can account for this. First, pedestal leakage into the 1 PE amplitude window is more pronounced at quadrant level, where four tiles are summed and the combined noise broadens the pedestal, pushing more counts into the counting region. Second, the vPDU test stand is substantially larger than the single-tile cryogenic setup, holding more than ten times the LN2 volume, hence any residual light produced internally by Cherenkov radiation, or externally through imperfect sealing, will affect the quadrant measurement more than the tile measurement. Both effects are expected to raise the quadrant

PCR relative to the tile sum.

Together, the anomalously low Oxford PCR relative to the production vPDU mean and the discrepancy between tile-average and quadrant rates triggered the subsequent investigation.

4 Results and Discussion

4.1 Comparison across Facilities

The analysis begins with available Grade A data from the vPDU testing laboratories in Edinburgh, Liverpool and AstroCeNT, followed by a comparison with our Oxford pre-production data. In spite of differences in the individual test stands, the reconstruction procedure is mostly similar, and the evaluation of the reconstructed data is consistent across facilities. As such, we run the passport characterisation on multiple vPDU quadrants and verify the PCR. Appendix E contains the initial PCR statistics, with tabulated values obtained for particular vPDUs in each location.

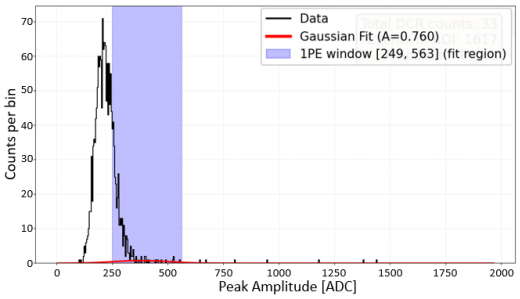
Inspection of the signal histograms from each facility follows, emphasizing a strong difference to data we collected and analysed in Oxford. The full amplitude spectra on log scale for selected test runs are shown in pairs in Figure 9. One immediate difference we notice is the significantly decreased SPE peak separation as opposed to Oxford. This is due to hardware differences in our pre-production vPDU, having a higher LV supply compared to those in the other facilities (2.5 V as opposed to 2.3 V). The discrepancy elsewhere is most evident in the 1 PE region, where a considerable portion of the pedestal is merged with the 1 PE peak. This is especially visible in tests from Liverpool, where the separation between the 0 PE and 1 PE peaks is the lowest, and least significant in Oxford, where almost no pedestal is counted. Inevitably, the pedestal leakage affects all time intervals, including those in the pre-trigger region used for measuring the PCR.

The issue with DN counting traces back to the passport artefact of including the pedestal leakage in 1 PE window signals, leading to an upward bias on PCR where the 0 PE and 1 PE peaks are insufficiently resolved. We implement a revised calculation method in order to evaluate the magnitude of this contribution to the PCR across laboratories, by considering not only the DN ROI but the entire pre-trigger region.

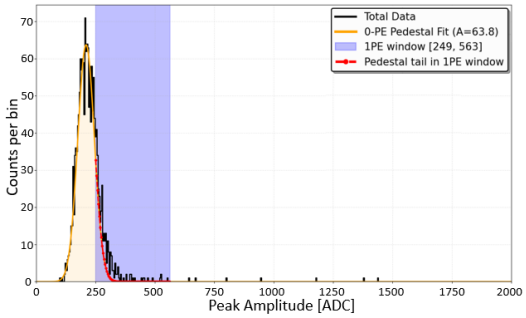
4.2 Corrected PCR

Three methods are investigated to remove the pedestal leakage into the PCR estimate, all based on Gaussian fitting. For the purpose of comparing methods, Q4 from vPDU120 in Edinburgh is used.

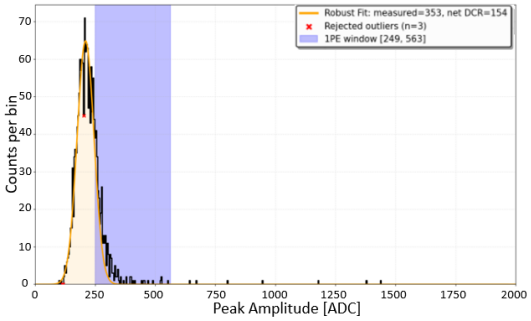
Gaussian fit for the 1 PE peak – This first method aims to fit a Gaussian function to the 1 PE region and only count the bins below it. Any attempt of directly



(a) 1 PE Gaussian fitting



(b) Pedestal Gaussian fitting



(c) Pedestal Gaussian fitting subtracting outliers

Figure 10: Strategies implemented for revising PCR measurements, implying Gaussian fitting and subtraction, conducted across all facilities, and illustrated for vPDU120 Q4.

fitting to the counts in the window is futile, given the ~ 10 statistics for each test. Figure 10a presents how such an approach yields an unreasonably low amplitude, under-counting even true 1 PE signals, with the red fit line appearing almost flat. Various other attempts to normalise the 1 PE fit were tried, however because the 1 PE statistics were so low the alternate strategy of fitting the pedestal and subtracting it from the 1 PE region was ultimately chosen.

Gaussian fitting and subtraction of the 0 PE peak – Given the more evident Gaussian shape of the pedestal, we proceed to consider a best fit on the 0 PE region, then count the bins below it which leak into the window, and finally subtract those counts from the total number computed originally. This process is displayed in Figure 10b, with the pedestal fit outside the window in yellow, and the subtracted tail in red. As can be seen, although the fit is more representative of the counts,

parts of the tail inevitably get omitted, still altering DN measurements. This is mainly due to bins such as those in the centre of the pedestal, which have lower counts than expected, drawing the fit down. An alternative method is tested here, once again artificially raising the amplitude to improve the subtraction, but it proves inconsistent across runs.

Pedestal fitting with subtracted outliers – The third and final method avoids amplitude manipulations and counts on χ^2 minimisation, while being consistent for all vPDUs. By simulating Poisson errors on each bin giving $\sigma = \sqrt{N}$, we impose a 2σ clipping of outliers, marked with red crosses, then recalculating the fit without them, shown in yellow in Figure 10c.

The final cleaned-up method is an upgraded version of this, applying a pre-defined function used in the QA/QC for fitting, which only considers portions of the curve which look Gaussian and also ignores outliers. The results for the same quadrant are provided in Figure 11 and a comparison of the initial and final methods is available in Table 2. The final method gives the largest difference in the newly estimated PCR from the initial value, presenting an over 4× reduction[6].

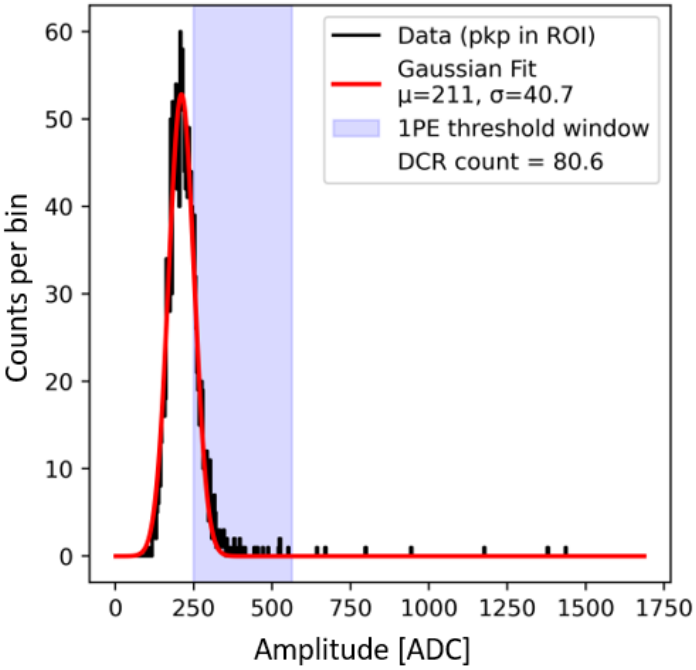


Figure 11: Gaussian-Subtracted DCR (GSDCR) approach for evaluating the PCR. It presents the best fit by a large margin, minimising false pedestal counts and significantly improving passport calculations across all facilities.

This method was deemed reliable and implemented in the main vPDU characterisation pipeline for a rerun of all previous tests. The new collected statistics are reported in Figure 12, and the initial method global results are provided in Table 3. More generally, the reduction

[6]Note that this is among the higher-PCR quadrants from Edinburgh, $\sim 1\sigma$ away from the initial QA/QC mean (Appendix D)

Table 2: Comparison of the different fitting methods for DN counting and the resulting PCR values for vPDU120 Q4.

Method	DN counts	DN Rate	PCR (Hz/mm^2)
Initial	345 ± 19	0.00690 ± 0.00037	0.779 ± 0.042
Final	81 ± 9	0.00160 ± 0.00048	0.182 ± 0.054

across facilities in terms of quadrant PCR is 2.7× for Liverpool (from 0.52 to 0.19 ± 0.01 Hz/mm^2), and 1.4× for AstroCeNT (from 0.50 to 0.36 ± 0.02 Hz/mm^2). The improvement is more significant for Liverpool data, with poor peak separation, and less so for AstroCeNT. A rerun of our initial Oxford pre-production vPDU tests leaves the results unchanged, confirming no overcorrection for large peak separation.

Tile-averaged PCR presents a 15% decrease to 0.14 Hz/mm^2 in Liverpool vPDUs, yet a 50% increase for AstroCeNT to 0.30 Hz/mm^2. This unexpected increase, as well as the remaining inter-facility difference in quadrant PCR, stem from the initial analysis including more low PCR tiles for AstroCeNT, in a full sample double in size than ours. Actual decrease for individual vPDUs is evident when comparing with the initial statistics in Appendix E[7]. The gap between quadrant and tile-average means was also reduced, from 3.2× discrepancy to 1.4× for Liverpool, respectively from 2.4× to 1.2× difference for AstroCeNT.

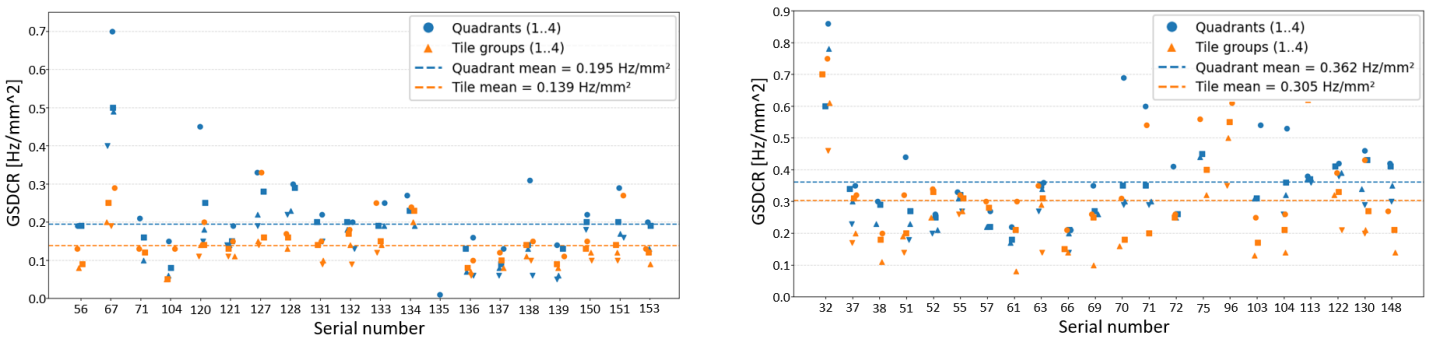
Table 3: Results of the tile average and quadrant PCR across facilities, computed with the initial method, prior to Gaussian Subtraction correction. The full initial statistics can be found in Appendix E.

Location	Quadrant-averaged vTile Mean (Hz/mm^2)	Quadrant-averaged PCR (Hz/mm^2)
Liverpool	0.16	0.52
Edinburgh	0.16	0.63
AstroCeNT	0.21	0.50

4.3 Dark Box Study and Error Analysis

Further tests were conducted at Oxford to investigate signal addition and reducing the incidence of spurious light onto the PDU during the PCR measurement. This second stage of testing benefited from an improved setup, consisting of an additional dark box enclosing the vPDU inside the dewar, providing substantially improved light isolation. The main effect of such a sealed enclosure is the elimination of residual external backgrounds, such as any ambient light still entering the vessel, or Cherenkov radiation from cosmic muons. Although LN2 was used instead of LAr, resulting in a lower scintillation rate[3], a meaningful contribution of scintillation light background is still expected. Any reduction in DCR between initial

[7]For instance, the high-PCR vPDU 37 was reduced from an average of 0.95 to 0.25 Hz/mm^2 with Gaussian Subtraction.



(a) Liverpool data **(b)** AstroCeNT data

Figure 12: Revised PCR measurement statistics across the Liverpool and AstroCeNT facilities, using the newly implemented GSDCR calculation.

and final tests is expected to be attributed to the elimination of these external sources, rather than a change in the SiPM dark noise having occurred in the meantime. Lastly, we modify half of the layout of the vPDU (Q1 and Q4) with four Grade A tiles and four with noticeably higher PCR than DCR (with values $\sim 0.001\,\mathrm{Hz/mm^2}$), to test both of the aforementioned properties. Final results are summarised in Table 4.

Table 4: Comparison of our measured PCR, GSDCR, and tile-average values from the dark box tests[8].

Test	PCR (Hz/mm^2)	GSDCR (Hz/mm^2)	Average (Hz/mm^2)
Q1	0.321 ± 0.008	0.027 ± 0.011	—
T1	0.217 ± 0.006	0.018 ± 0.008	0.016 ± 0.008
T2	0.196 ± 0.006	0.014 ± 0.008	
T3	0.223 ± 0.006	0.017 ± 0.008	
T4	0.207 ± 0.006	0.015 ± 0.008	
Q4	0.342 ± 0.008	0.025 ± 0.011	—
T13	0.232 ± 0.006	0.012 ± 0.009	0.013 ± 0.008
T14	0.215 ± 0.006	0.017 ± 0.008	
T15	N/A	N/A	
T16	0.200 ± 0.006	0.010 ± 0.008	

We report a decrease in GSDCR from initial Oxford tests with this new counting method, coming as the result of the dark box setup, which no other runs used. Specifically, we obtain average GSDCR values of (0.063 ± 0.026) and (0.047 ± 0.021) Hz/mm^2 for Q1 and Q4 tiles respectively, in the absence of the dark box, compared to the (0.016 ± 0.008) and (0.013 ± 0.008) Hz/mm^2 with the dark box. Because of the light isolation, the standard passport fails to initially identify at least two resolved peaks in the laser ROI and run the analysis with the PCR – to mitigate this, the full-waveform spectrum is considered for the peak identification, along with a reduction of the peak-finding sensitivity, so we are able to analyse the low-statistics regime as well with our GSDCR method[9]. Comparing tiles and quadrants, we remark rates $1.5 - 2\times$ higher for quadrant measurements than the corresponding tile averages even with light reduction – this is still an open question.

As a validation of the GSDCR method's sensitivity, a blind prediction is performed. Without knowledge of which quadrant contained the anomalous tiles, our expectations relating to signal uniformity suggest that Q1, containing the more uniformly spread GSDCR of tiles and closer average to the quadrant value, should contain the *good* tiles, while the corresponding behaviour in Q4 marks the presence of abnormal ones. Our predictions are verified, and the measurements for the *good* tiles are found to be consistent with the tile test stand dark box PCR, of $0.011 - 0.015\,\mathrm{Hz/mm^2}$. The order of PCR values in these tiles is also preserved with T2 < T4 < T3 < T1 (see Appendix D for the full details).

For the original counting method, the PCR uncertainty follows directly from Poisson counting statistics. Thus, $\overline{N}_{\mathrm{DN}}$, the mean of hit counts per waveform $\{n_i\}$, has an uncertainty given by the standard error of the mean

$$\sigma_{\overline{N}} = \frac{\mathrm{std}(\{n_i\})}{\sqrt{N_{\mathrm{WF}}}}, \tag{6}$$

which propagates to the PCR as $\sigma_{\mathrm{PCR}} = \sigma_{\overline{N}}/(T_{\mathrm{ROI}} \times A)$. The intermediate approaches assume Poisson errors on bin counts, $\sigma_i = \sqrt{N_i}$, which further affect DN rates. As for the final GSDCR, the error is calculated from the Poisson combination of the total measured counts and the estimated pedestal overlap within the 1 PE window:

$$\sigma_{N_{\mathrm{corr}}} = \sqrt{N_{\mathrm{total}} + N_{\mathrm{overlap}}}\,. \tag{7}$$

Here, $N_{\mathrm{total}} = \sum_{i \in 1\ \mathrm{PE}} N_i$ is the total count in the 1 PE window and $N_{\mathrm{overlap}} = \sum_{i \in 1\ \mathrm{PE}} \min(N_i,\, N_i^{\mathrm{ped}})$, the

[8] In the current analysis, tile 15 was completely off, and was excluded from the data. This observation is consistent with us obtaining ∼462,000 total counts for the Q1 channel and ∼329,000 for the Q4 channel, suggesting an approximately 4/3 ratio given by the number of active tiles.

[9] Instead of directly fitting the laser ROI finger plot, which contains too few 1 PE counts to resolve peaks with the dark box light suppression, the 1 PE position is extracted from a coarse histogram of all amplitudes across the full waveform. This provides a valid threshold estimate with as few as ∼ 10 counts above the pedestal.

estimated pedestal contribution (N_i^{ped} being the predicted pedestal count in the ith bin). The normalisation of the pedestal fit is extracted from the laser ROI, by scaling that count density per bin width to the width in the pre-trigger region.

Apart from statistical uncertainties, we must consider a number of systematic sources of error. The laser pulse produces an excess of counts near the boundary of the 0 PE peak at the trigger time, caused by light scattered inside the dewar that reaches the SiPMs and produces low amplitude pulses that alter the pedestal shape; this implies the pedestal is not perfectly Gaussian itself, perpetuating uncertainty in all fitting and subtraction attempts[10]. The 3σ peak-finding threshold is carefully set to optimise SNR for the Oxford test stand, yet it may not transfer perfectly for runs with different baseline RMS levels. Within this section, presenting Oxford dark box tests only, the dominant systematic is residual light leakage that varies between runs depending on the sealing conditions[11].

5 Conclusions and Further Analysis

We have presented a characterisation of the Pulse Count Rate in veto Photo-Detector Units for the DarkSide-20k experiment, resolving a disagreement between Oxford measurements and QA/QC references, while implementing a revised counting procedure beneficial across all testing sites.

Initial Oxford PCR measurements of $\sim 0.06\,\text{Hz/mm}^2$ were approximately one order of magnitude below the previously well-established mean. This suggested inconsistencies in the signal reconstruction procedure, and were traced to leakage of the pedestal into the amplitude region of interest used for the PCR calculation in the vPDU passport. The computation pipeline was manually validated and a methodical inter-facility comparison identified significantly higher pedestal leakage into the DN measurement window, for tests conducted in Edinburgh, Liverpool and AstroCeNT. This reflects the difference between production and pre-production hardware, primarily by a 0.2 V lower LV supply at those sites, which reduced peak separation and motivated the development of an improved calculation method. We created a Gaussian-subtracted dark noise correction, which was implemented and reduced initial erroneous external PCR values by 2.7× for Liverpool and 1.4× for AstroCeNT. The discrepancy between tile-averaged and quadrant PCR was also improved, from 3.2× to 1.4× for Liverpool, respectively from 2.4× to 1.2× for AstroCeNT.

Subsequent testing with an internal PDU dark box setup confirmed that external photon backgrounds are the principal effect elevating PCR and signal addition analysis exhibited $1.5-2\times$ excess PCR in quadrant measurements than the equivalent tile average. *Good* and biased tile behaviour was studied and accurately predicted from values and patterns of the newly calculated rates. A parallel defect correlation study on 39 tested and inspected tiles also revealed that certain electrical and mechanical faults are modest indicators of raised PCR, $60-70\%$ of tiles in these categories surpassing the global median.

One natural next step for this analysis would be to fit the full high statistics outside-ROI data from production testing with a multi-PE fit, to avoid doing additional corrections. Analysis of the inclusion of light, which varies from one test stand to another depending on geometry, would be useful as well, so one could perform a calibration in each test stand of how much the PCR changes with vs. without the internal dark box and obtain a calibration factor. Extending the account of uncertainty propagation in the Gaussian subtraction method would be a plus, however the unpropagated fit error is relatively small compared to the statistical error on our ~ 100 events for the PCR calculation. Examining a larger sample for the defect correlation study, such as to consider multiple tiles with similar defects and improve the statistics for each class or consider more specific subclasses, would greatly benefit us. Equally relevant, performing dark box tests across all facilities would allow for the differentiation of light background and systematic sources, providing a valid reference for the 120 vPDUs awaiting installation.

[10] Subsequent approaches of fitting the full amplitude spectrum on log scale for the dark box runs are presented in Appendix C, where the offset from a Gaussian in the pedestal is clear.

[11] Residual light is suppressed by the dark box, but not completely eliminated, with different runs for tiles and quadrants being conducted with potentially different light levels and tape sealing which degrades over time.

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[7] S. Gundacker and A. Heering. The silicon photomultiplier: fundamentals and applications of a modern solid-state photon detector. *Physics in Medicine and Biology*, 65:17TR01, 2020.

A vPDU Installation and Testing

vPDU disassembly and reassembly involves a careful procedure, involving special trays, fixtures and supports. For the final tests we conducted, we changed eight of the total 16 tiles of vPDU3, which can be seen fully connected in the pictures below. We also present a specific instance in the data acquisition process, where we analyse IV curves and ramp up the voltage to analyse photo-electron signals on the oscilloscope.

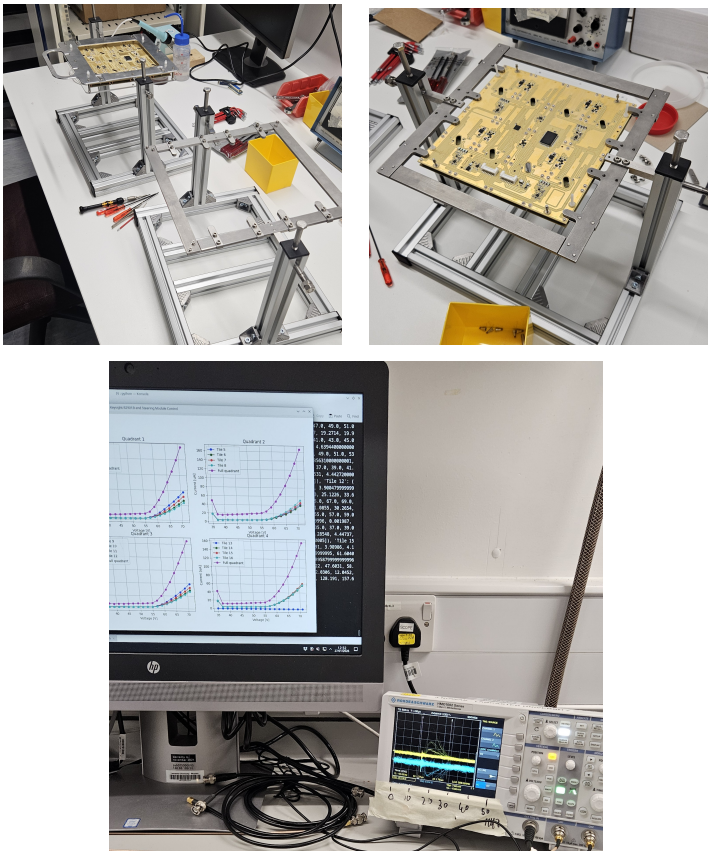


Figure 13: (Top) vPDU installation setup, before and after instances. (Bottom) IV curve taking, and data acquisition, which can be visualised on the oscilloscope, with the differently sized peaks on each channel representing signals of 1 PE, 2 PE etc., in decreasing order of intensity according to their prevalence in the spectrum.

B Reconstruction and Analysis Code

The relevant code fragments used throughout this project can be accessed via github, including the final approaches constructed on top of the initial procedure: **Revised Reconstruction and Analysis**

C Gaussian fitting to log spectrum

For the Gaussian Subtraction method implemented, we always used fitting on the pre-trigger ROI to directly extract DN rates from there. However, an interesting attempt of ours also involved fitting the full logarithmic finger plots, which is computationally easier to do, especially in low statistics regimes like those generated by dark box tests, and may be implemented in future iterations. One such fit of the 0 PE and 1 PE peaks for the Q4 data of our final tests is illustrated in Figure 14.

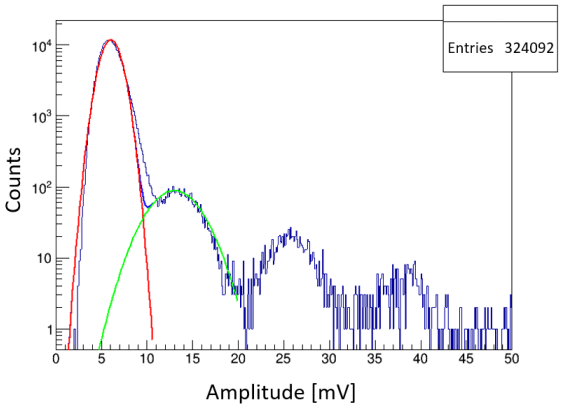


Figure 14: Further attempts at Gaussian fitting of log finger plots for the abnormal tiles in Q4 (with the dark box setup).

D Initial Oxford results and QA/QC study

We further provide a number of tables referenced throughout the report, which prove the accuracy of our data and interpretation. Table 5 represents the eight tiles we reinstalled and tested with the dark box. It shows initial measurements of PCR values and we can clearly see the spread of results in the quadrants, as well as the order in quadrant 1 consistent with our dark box results.

Table 5: QA/QC PCR values for selected tiles and their equivalent positions in the vPDU.

Tile number	in vPDU	QA/QC PCR (Hz/mm^2)
1731	T1	0.1470
782	T2	0.1120
1656	T3	0.1460
3005	T4	0.1440
1744	T13	0.0006
1809	T14	0.0070
916	T15	0.1810
2120	T16	0.0042

Table 6 highlights the initial results the passport yielded for Q1 of vPDU3. The low PCR and CDA values are evident in comparison to QA/QC results, shown in Figure 15, and valid for quadrants. Also for quadrant tests, we note that NAPA and SNR values are well within the QA/QC boundaries, justifying our further inspection of the analysis procedure and not directly claiming hardware defects are the root of this issue.

E Initial cross-facility statistics and comparison table for Grade A tiles

The initial statistics from the Edinburgh, Liverpool and AstroCeNT laboratories are summarised in Figure 16.

F Full defect correlation study

The detailed analysis of the 39 tiles, ordered by decreasing PCR, is shown in Figure 17.

Table 6: Summary of initial run results and manually found entries in the DN ROI. The passport failed to fit the peaks for tile 2 in this configuration, yielding NaN values – this was later corrected in the dark box analysis.

Run	**Data**	$N_{\mathbf{WF}}$	**DN Rate**	**PCR (Hz/mm^2)**	**CDA**	**NAPA**	**SNR**	**#ROI**
376	Q1	309000	5.02×10^{-4}	0.057	0.211	0.57 ± 0.16	5.25 ± 0.04	154
377	Q1T1	312165	8.97×10^{-5}	0.041	0.188	0.98 ± 0.12	9.25 ± 0.04	29
378	Q1T2	309000	nan	nan	nan	2.71 ± 1.80	nan	29
379	Q1T3	305000	7.86×10^{-5}	0.036	0.191	1.04 ± 0.21	10.80 ± 0.03	24
380	Q1T4	318000	1.23×10^{-4}	0.055	0.191	1.50 ± 0.52	9.40 ± 0.03	40
381	Q1T1T2	313000	1.56×10^{-4}	0.035	0.197	1.32 ± 0.15	6.94 ± 0.03	51
382	Q1T1T4	306000	1.70×10^{-4}	0.038	0.199	1.38 ± 0.21	7.56 ± 0.03	54
383	Q1T1T3	306000	1.47×10^{-4}	0.033	0.197	1.49 ± 0.21	6.98 ± 0.03	46

vPDU Passport Summary

--- Mean

--- Mean Â±3 sigma

--- QA/QC limits (Grade A)

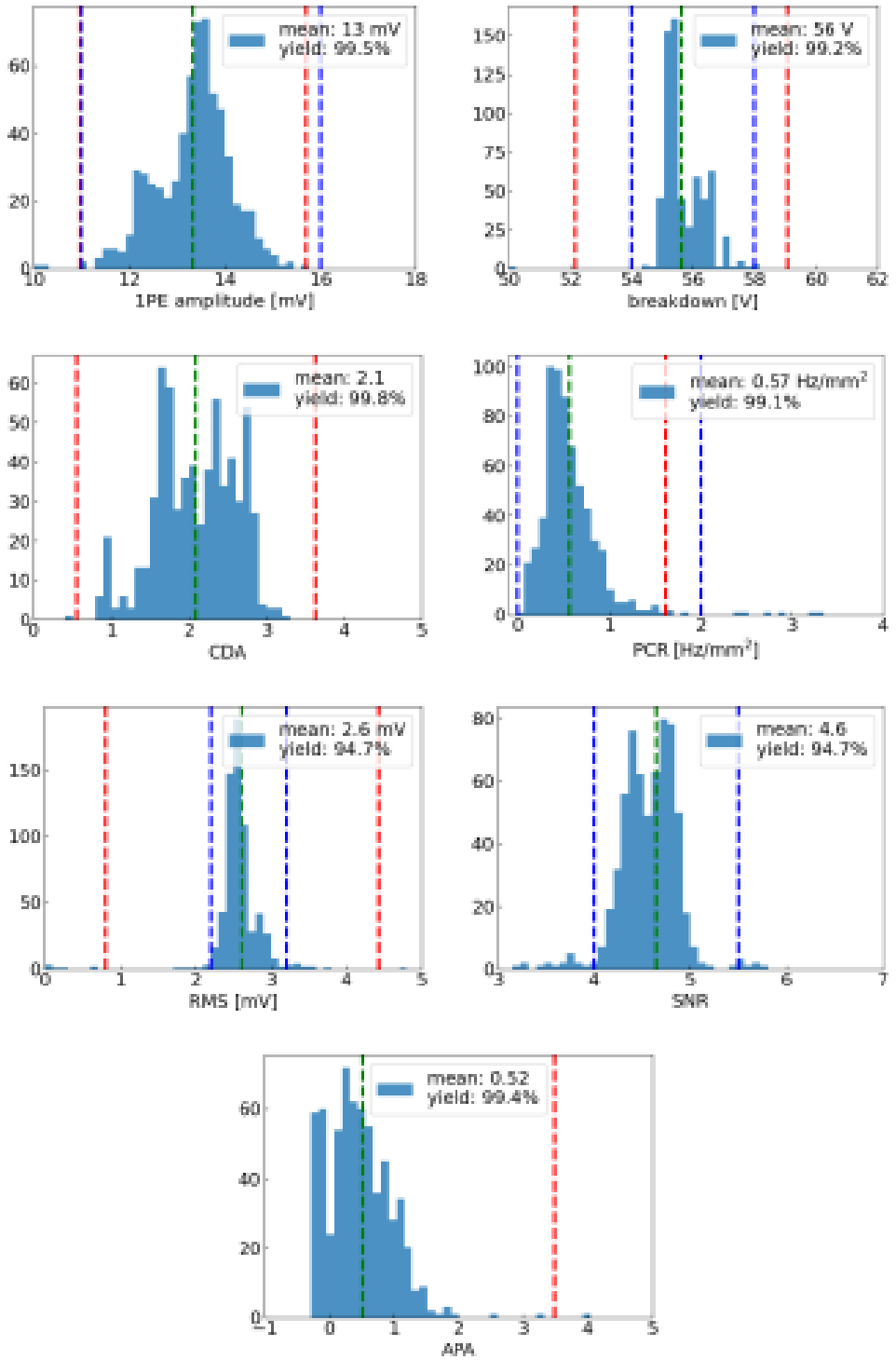
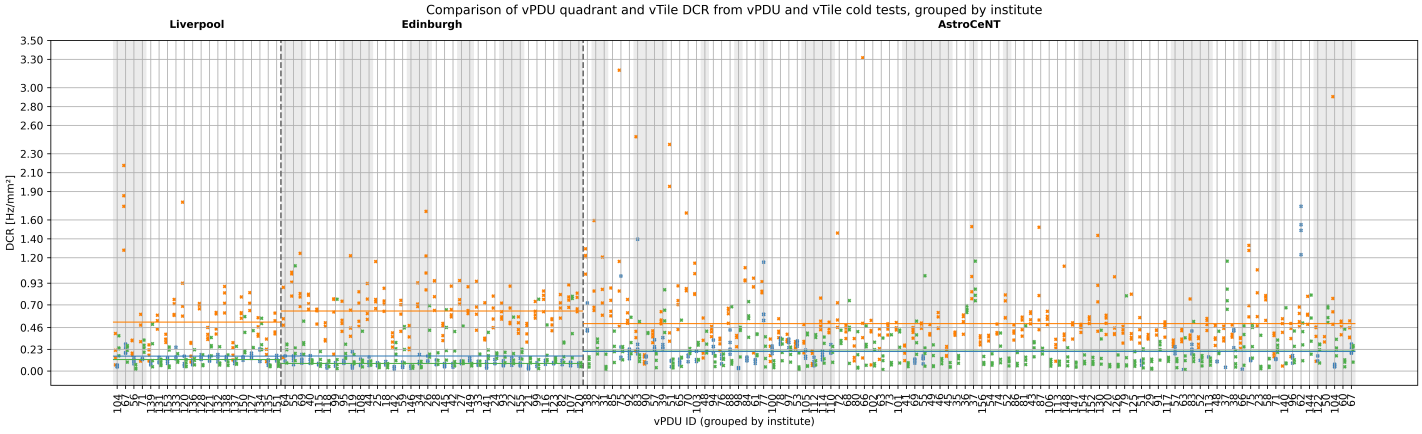


Figure 15: Full initial QA/QC study, with acceptance margins and averages indicated. CDA and APA are outputs of the passport and useful indicators, yet not part of the QA/QC parameters, as can be seen from the absence of the blue Grade A thresholds on the corresponding plots.



	Quadrant-averaged vTile Mean (vTile cold test)	Quadrant-averaged vTile Mean (vPDU cold test)	Quadrant DCR
Liverpool	0.12	0.16	0.52
Edinburgh	0.08	0.16	0.63
AstroCeNT	0.21	0.21	0.5

Figure 16: Initial statistics of tile and vPDU DCR tests from Liverpool, Edinburgh and AstroCeNT. We note the relevant results for our study, which are the quadrant-averaged tile mean PCR of 0.16, 0.16 and 0.21 Hz/mm^2, and the quadrant-averaged PCR of 0.52, 0.63 and 0.50 Hz/mm^2 for the three facilities, respectively. The latter are 3.2×, 3.9× and 2.4× larger than the former in this initial study. These results are also described in Table 3 of Section 4.2.

Table 7: Individual test run data of Grade A quadrants from different facilities, highlighting the order of magnitude difference between external tests of production PDUs and the preliminary one we conducted in Oxford on our different, pre-production PDU (final row).

Site	Run	vPDU	Quadrant	1 PE lower (mV)	1 PE higher (mV)	Passport DN Rate
Liverpool	1029	56	1	8.97	19.28	0.0028
			2	8.42	18.54	0.0029
			3	8.98	18.57	0.0017
			4	9.19	19.07	0.0015
	1029	104	1	8.28	18.86	0.0020
			2	8.03	20.26	0.0035
			3	8.84	20.08	0.0012
			4	9.17	18.74	0.0006
Edinburgh	2699	107	1	8.04	17.28	0.0073
			2	7.90	17.40	0.0058
			3	8.05	17.40	0.0056
			4	7.60	17.16	0.0069
	2665	21	1	7.79	16.79	0.0045
			2	8.29	17.36	0.0035
			3	8.12	17.96	0.0043
			4	8.32	18.02	0.0027
	2620	24	1	8.13	17.07	0.0070
			2	8.47	17.23	0.0049
			3	7.66	16.68	0.0059
			4	8.08	16.87	0.0041
	2528	146	1	8.04	17.08	0.0032
			2	7.59	17.00	0.0043
			3	7.86	16.84	0.0020
			4	8.27	17.62	0.0021
	2596	149	1	8.02	17.07	0.0055
			2	7.99	16.92	0.0054
			3	7.44	16.37	0.0079
			4	8.12	17.27	0.0057
	2700	120	1	7.99	17.40	0.0054
			2	7.99	17.27	0.0058
			3	8.46	17.24	0.0043
			4	7.61	16.99	0.0066
AstroCeNT	844	88	1	8.79	17.85	0.0059
			2	8.71	17.88	0.0054
			3	9.08	18.49	0.0045
			4	8.65	18.41	0.0057
	844	94	1	9.15	18.66	0.0040
			2	9.41	18.77	0.0032
			3	9.05	18.66	0.0037
			4	9.12	18.35	0.0042
Oxford	376	3	1	9.95	18.32	0.0005

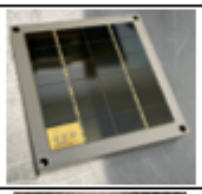
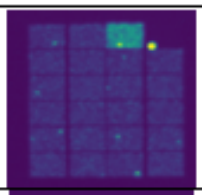
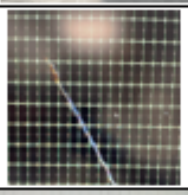
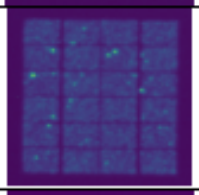
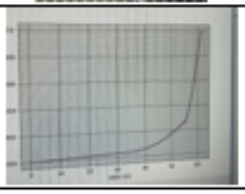
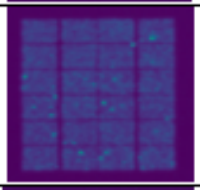
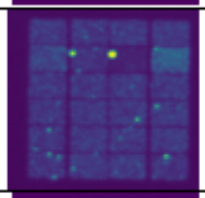
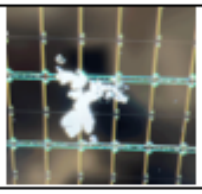
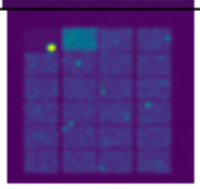
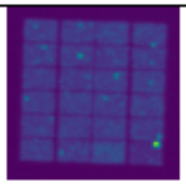
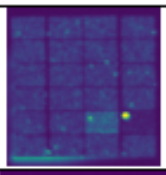
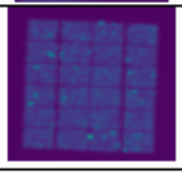
Tile	DCR (Hz/mm^2)	Class	CMOS Image	Defect Image	Defect Note
794	1.353620	failure	N/A		current limit; missing SiPM; second rework failed warm
450	0.715173	failure			double BD; surface damage
820	0.538260	failure			high RMS; failed cold IV; double BD on retest
1303	0.512987	electrical		N/A	high PCR; quadrant BAD
2059	0.483200	mechanical			extra SPE peak + high RMS; dirty tile
565	0.423026	unknown		N/A	no surface damage noted; defect not specified
104	0.407682	mechanical			bad SPE; good quadrant; broken wire bond, bent bonds, deposits
1732	0.301173	electrical			extra SPE peak; no visible damage; detached fixture
1751	0.289740	electrical		N/A	high RMS + high PCR; quadrant BAD

Figure 17: Detailed analysis of the defect correlation study conducted on 39 tiles with known PCR (part 1 of 4).

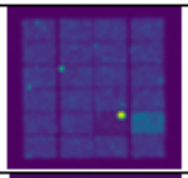
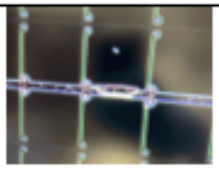
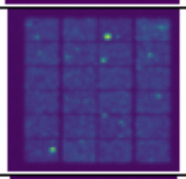
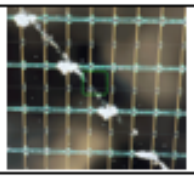
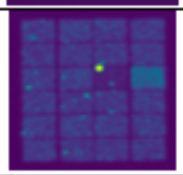
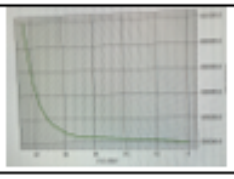
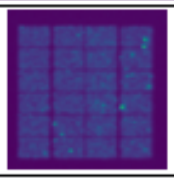
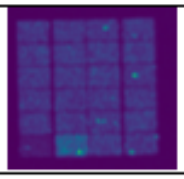
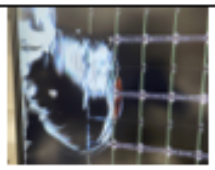
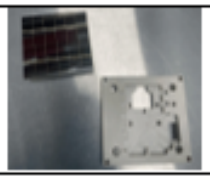
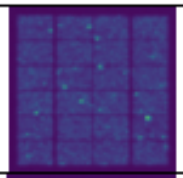
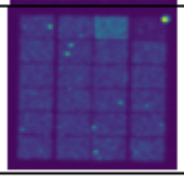
1271	0.285527	mechanical			bad SPE; good quadrant; small rail damage
1894	0.276501	mechanical			high PCR; surface damage
2147	0.243706	electrical		N/A	high PCR; no surface damage
1185	0.213619	failure			current limit; breakdown ~35V; wire bond damage
609	0.157055	electrical		N/A	high RMS + high PCR; quadrant BAD
416	0.151639	failure	N/A	N/A	no LV / ASIC issue; no CMOS
304	0.129074	mechanical			deposit by damaged SiPM
747	0.123959	mechanical	N/A		surface damage; fell face-first onto fixture
778	0.117039	electrical		N/A	high PCR; retest Grade A
733	0.113128	mechanical			extra SPE peak; surface damage

Figure 17: Detailed analysis of the defect correlation study (part 2 of 4).

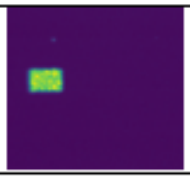
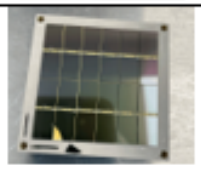
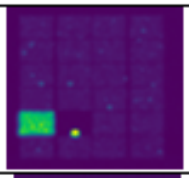
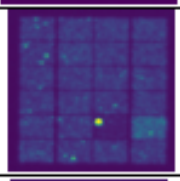
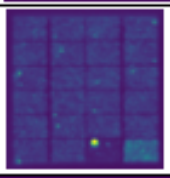
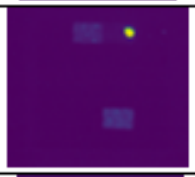
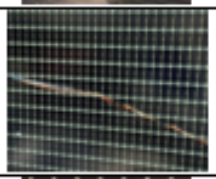
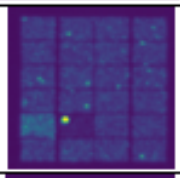
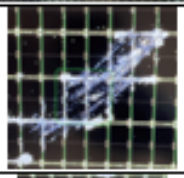
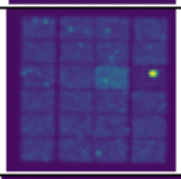
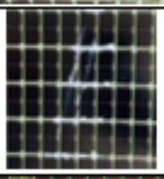
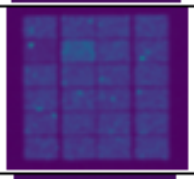
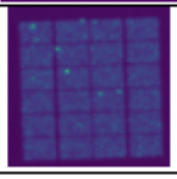
1246	0.111323	failure			current limit; breakdown ~35V; wire bond damage
326	0.092067	mechanical	N/A		shipping fixture with broken SiPM; risk damaged
586	0.092669	electrical		N/A	bad SPE; no surface damage
910	0.090562	electrical		N/A	high PCR; no visible damage
167	0.087253	mechanical			extra SPE peak; surface damage
1708	0.097483	failure			double BD; big scratch; current limit ~40V
2150	0.062281	mechanical			high PCR; surface damage
302	0.056564	mechanical			high PCR; surface damage; 2nd rework failed cold
137	0.050847	failure			double BD; bent wire bonds; scratches
331	0.043025	electrical		N/A	high RMS + high APA; retest Grade A; no bad SiPM

Figure 17: Detailed analysis of the defect correlation study (part 3 of 4).

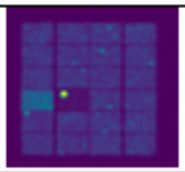
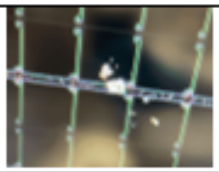
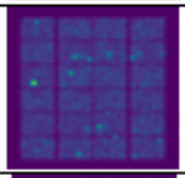
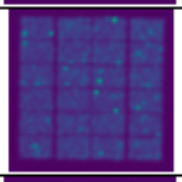
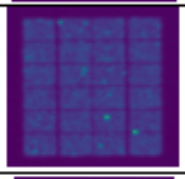
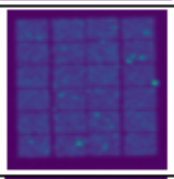
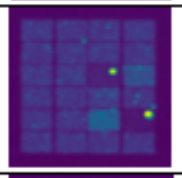
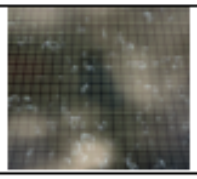
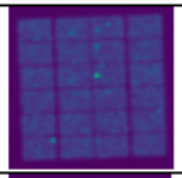
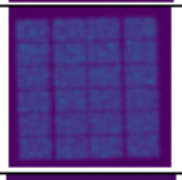
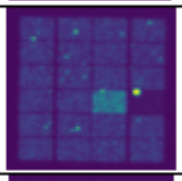
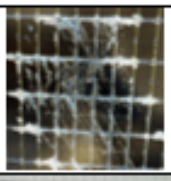
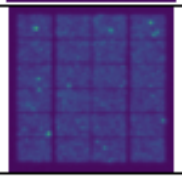
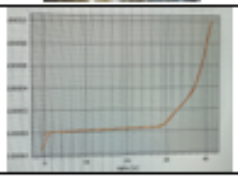
1304	0.038211	mechanical			bad SPE + high APA; small rail damage
1452	0.036105	electrical		N/A	high PCR; quadrant BAD
2080	0.028583	failure		N/A	current limit; tile OFF
1307	0.028884	electrical		N/A	bad SPE; quadrant BAD
1288	0.025273	electrical		N/A	bad SPE; quadrant BAD
369	0.024371	failure			double BD; scratches and detritus
788	0.020158	unknown		N/A	not specified in text extract
517	0.017751	unknown		N/A	not specified in text extract
519	0.016247	mechanical			extra SPE peak; large SiPM damage
341	0.012637	failure			current limit; failed cold IV → double BD on retest

Figure 17: Detailed analysis of the defect correlation study (part 4 of 4).